

ZACKARY LOEVSETH

Independent Researcher | Human-AI Systems | Computational Mathematics | Scientific Infrastructure

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PERMANENT PORTFOLIO: <https://zackaryloevseth.github.io/research-portfolio/>

PROFESSIONAL SUMMARY

Independent researcher and research-program director working at the intersection of human-AI systems, mathematical research, decision and provenance theory, and scientific reproducibility. Creator of the Theory of Epistemic Control in Human-AI Systems and Revision-Robust Epistemic Transition Integrity (RR-ETI). Leads multi-model, tool-assisted research from target selection through proof reconstruction, exact computation, hostile audit, reproducibility, and publication staging. Current portfolio includes three scientifically publication-ready mathematics manuscripts, exact computational certificate archives, and a preregistered validation program.

RESEARCH LEADERSHIP

Independent Researcher & Research Program Director | Theory of Epistemic Control / RR-ETI

2026-Present

- Formulated RR-ETI as a finite decision-and-provenance framework for preserving and repairing materially dependent claims after noisy premise invalidation; developed exact additive-model results and a closure-constrained research frontier.
- Built an operational epistemic runtime separating generation, verification, authorization, execution, publication, and completion; implemented typed claim ledgers, correction propagation, independent validation, versioned checkpoints, and recoverable handoffs.
- Designed Study 14 as a living validation program and Study 14.1 as a four-condition comparison of context and premise-governance strategies; preserved confirmatory boundaries while instrumenting naturalistic research campaigns.
- Directed a portfolio-scale autonomous scientific workflow that moved three distinct mathematical campaigns to publication-ready scientific packages while retaining human control over authorship, external communication, and final publication.

PUBLICATION-READY PREPRINTS (NOT YET SUBMITTED)

Explicit Counterexample Families to Five Finite-Group Inequalities from Graffiti3 | Finite group theory

2026

- Established explicit infinite counterexample families: $U7(q)$ for three inequalities and dihedral families for two additional inequalities; retained compact witnesses $U7(2)$, $D64$, and $D50$.
- Directed a versioned GAP SmallGrp census through order 256, exact arithmetic and implementation-independent checking, mutation testing, and a complete reproducibility package.

Largest-Index Zonotope Bounds for Planar Projections of Semiorder Ideals | Combinatorial optimization

2026

- Derived instance-sensitive upper- and full-hull bounds in terms of incomparable pairs, together with $O(nr)$ complexity for represented semiorders of width r and a mixed-sign linear lower construction.
- Validated the implementation with a source-independent checker covering 318,667 cases, 3,978,057 ideals, and 2,311,018 projected points; corrected and replayed a zero-mutation reproducibility archive.

Excess-Degree Bounds for Minimal Counterexamples to the Erdős-Gyárfás Conjecture | Graph theory

2026

- Proved the excess-sensitive bound $|A| \geq 2b+s+4$ for lexicographically minimal hypothetical counterexamples, with density and edge-count consequences; preserved that the global conjecture remains open.
- Coordinated definition-level proof reconstruction, hostile audit, two method-distinct implementations, 21 publication gates, and a clean-extraction reproducibility package.

ADDITIONAL RESEARCH PROGRAMS

Erdős Problem #409 - Certified Record Computations | Number theory / exact computation 2026

- Directed a public research archive certifying record trajectories through $F=71$ for iteration of $n \rightarrow \phi(n)+1$, including 72 trajectory nodes, complete factorizations, and 222 recursive Lucas/Pratt-style prime certificates.
- Required structurally distinct implementations, factor-from-scratch recomputation, deterministic serialization, SHA-256 manifests, continuous integration, and explicit separation of finite record results from global claims.

Riemann Hypothesis Research Program | Analytic number theory / certified computation 2026-Present

- Directed a target-preserving program that produced an internally checked rational-bump Weil reduction, computable bump bounds, a conditional finite-window Li-coefficient theorem, and a conditional uniform-computability architecture for coded Weil entries.
- Built producer-checker evaluation pipelines that rejected under-supported certificates and localized proof-transcript obligations; maintained the exact global status RH UNSOLVED / IN PROGRESS.

PROFESSIONAL EXPERIENCE

Canvassing Field Manager / Director | Fund for the Public Interest - Los Angeles 2022-2024 (documented)

- Advanced from Field Manager to Director, overseeing Los Angeles office operations and reporting performance metrics to regional and state leadership.
- Trained and managed 3-5 canvassers daily, led meetings, set goals, coordinated schedules and transportation, and participated in hiring, performance management, and team development.
- Recognized as a top-earning canvasser nationally after raising more than \$20,000 in one summer and exceeding the documented office average by approximately 500%.

Social Worker, Chef & Server | The Salvation Army - Bergen, Norway 2018-2022

- Prepared and served food for 30+ patrons, maintained kitchen inventory and budget needs, completed patron records, and advocated for people experiencing substance-use disorder.
- De-escalated conflicts and maintained safety in a high-pressure social-service environment.

Event Supervisor | FFF - Oslo, Norway 2020

- Constructed environmental-justice installations, provided educational resources and attendee support, and coordinated event security and logistics.

EDUCATION

B.S. Geography and Environmental Science; Minor in Geology | California State University, Northridge Status to confirm

- GPA: 3.8. Prior resume listed expected graduation in Fall 2024; update degree completion and graduation date before external use.

TECHNICAL & RESEARCH SKILLS

Research methods: theorem formulation, proof reconstruction, exact enumeration, computational graph theory, finite-group computation, combinatorial optimization, number theory, interval and rational arithmetic, experimental design. **Programming and tools:** Python, Ruby, C, Bash/Zsh, Git/GitHub, LaTeX, GAP/SmallGrp, Lean, FLINT/Arb, SymPy, clean-checkout replay, Cl. **Scientific infrastructure:** independent checkers, mutation testing, deterministic archives, SHA-256 manifests, provenance ledgers, Git bundles/worktrees, versioned checkpoints, publication staging. **Human-AI systems:** multi-agent research orchestration, prompt and contract design, claim-status governance, correction propagation, evaluator separation, resource-bounded autonomy, human-final publication controls.

LEADERSHIP & COMMUNITY

Vice President, Association of Norwegian Students Abroad - Los Angeles (2021-2023) | Young Friends of the Earth Norway (2018-2021) | Norwegian Union of Municipal and General Employees (2018-2021) | TreePeople volunteer (2023; current status to confirm)